



Male mammary gland development and methylation status of estrogen receptor alpha in Wistar rats are modified by the developmental exposure to a glyphosate-based herbicide

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ABSTRACT

Postnatal treatment with glyphosate-based herbicides (GBHs) induces endocrine-disrupting effects on the male rat mammary gland. In this study, the effects of developmental exposure to GBH on mammary gland growth and development, and the possible molecular mechanisms involved, were evaluated in pre- and post-pubertal male rats. To this end, pregnant rats were orally exposed through the food to 0, 3.5 or 350 mg GBH/kg bw/day from gestational day 9 until weaning. Mammary gland development and estradiol (E2) and testosterone (T) serum levels of male offspring were evaluated on postnatal day (PND)21 and PND60. Besides, prolactin (PRL) serum levels, proliferation index, androgen (AR) and estrogen receptor alpha (ESR1) expression, ESR1 alternative transcript mRNA levels, and DNA methylation status of ESR1 promoters were assessed on PND60. No differences between groups were observed in mammary gland development at PND21 or in E2 and T levels on both PNDs studied. On PND60, GBH3.5-exposed animals presented similar mammary gland histology but higher AR protein expression than controls, whereas GBH350-exposed males presented a less developed mammary gland, accompanied by a lower proliferation index, similar AR levels, and slightly increased PRL serum levels than controls. In both exposed groups, ESR1 expression was lower than in control rats, being lower in GBH350-exposed rats. GBH also altered the abundance of ESR1 transcript variants by hypermethylation of ESR1 promoters. GBH3.5 decreased only ESR1-OS expression, whereas GBH350 affected ESR1-O, OT and E1 expression. Our results show that developmental exposure to GBH induces epigenetic changes in ESR1, which could be responsible for the altered male mammary gland development observed in GBH350-exposed animals.

1. Introduction

Glyphosate (N-phosphonomethyl glycine) is an active ingredient of broad-spectrum herbicide formulations, whose primary mechanism of action is the inhibition of 5-enolpyruvylshikimate 3-phosphate synthase, an enzyme essential for the formation of aromatic amino acids in plants (Steinrucken and Amrhein, 1980). The use of glyphosate-based herbicides (GBHs) has rapidly increased over the last two decades and is expected to continue to increase due to the emergence of glyphosate-

resistant weeds and increased applications of GBHs, such as for pre-harvest crop desiccation (Benbrook, 2016; Vandenberg et al., 2017). In Argentina, more than 200 million liters of GBHs are applied every year (Aparicio et al., 2013) and their extensive use has been accompanied by the constant increase in transgenic glyphosate-resistant soybean production (CASAFA, 2012). Recent monitoring studies have evidenced the presence of glyphosate residues and its metabolite aminomethylphosphonic acid (AMPA) in different regions of Argentina, such as Buenos Aires, Corrientes, Córdoba, Entre Ríos, La Pampa and Santa Fe. These

Abbreviations: AMPA, aminomethylphosphonic acid; AR, androgen receptor; ESR1, estrogen receptor alpha; E2, estradiol; F0, dams directly exposed to GBH; GBH, glyphosate-based herbicide; GD, gestational day; LD, lactation day; LN, lymph node; LOD, limit of detection; LOQ, limit of quantification; PND, postnatal day; PRL, prolactin; TEBs, terminal end buds; T, testosterone; WM, whole-mount

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compounds have been detected in surface water, sediments, and soil (Aparicio et al., 2013; Bonansea et al., 2017; Mac Loughlin et al., 2017; Primost et al., 2017; Ronco et al., 2016), as well as in the respirable dust emitted by agricultural soil (Mendez et al., 2017). Furthermore, these compounds have been detected in a variety of crops at harvest and in processed food (EFSA, 2017a; Myers et al., 2016), as well as in human urine samples (Connolly et al., 2017; Mills et al., 2017; Niemann et al., 2015; Parvez et al., 2018) and maternal and umbilical cord serum (Kongtip et al., 2017). This widespread presence of these compounds shows that there is a risk of environmental exposure and concern about their possible effects on the environment and human health. Several studies have reported adverse effects of GBH exposure on both the female and male reproductive systems at both low and environmentally relevant doses (Mesnage et al., 2015; Székács and Darvas, 2018). In adult female rats, it has been shown that GBH affects the expression of estrogen-sensitive genes in the uterus (Varayoud et al., 2017), whereas early postnatal exposure alters uterine development (Guerrero Schimpf et al., 2017), the reproductive performance and uterine signaling (Ingaramo et al., 2016, 2017) and enhances the sensitivity of the rat uterus to estradiol (E2) (Guerrero Schimpf et al., 2018). Also, perinatal exposure to GBH impairs the female reproductive performance and induces structural congenital anomalies in F2 offspring (Milesi et al., 2018). In male rodents, a systematic review has shown that glyphosate exposure decreases sperm concentration (Cai et al., 2017). In addition in rats, adult and perinatal exposure to GBH decreases testosterone (T) serum concentration, increases aromatase expression levels in the testis, and increases abnormal sperm morphology (Cassault-Meyer et al., 2014; Dallegrave et al., 2007; Owagboriaye et al., 2017). However, Romano et al. (2010) and Romano et al. (2012) showed that, depending on the window of exposure, the effects on steroid hormone serum levels are different. Gestational and early postnatal exposure to GBH increases T and E2 concentration (Romano et al., 2012), whereas peri-pubertal exposure to GBH decreases T but does not affect E2 levels (Romano et al., 2010). Also, it has been reported that glyphosate may induce human breast cancer cell proliferation by directly activating estrogen receptor alpha (ESR1) *in vitro* (Thongprakaisang et al., 2013) or by activating ESR1 through a ligand-independent mechanism (Mesnage et al., 2017). Despite all the above-mentioned experimental data, there is still controversy about the endocrine disrupting potential of glyphosate, considering that no evidence of the potential interaction of glyphosate with endocrine pathways has been detected by the United States Environmental Protection Agency (US EPA) (EPA, 2015) or the European Food Safety Authority (EFSA) (EFSA, 2017b).

Exposure to certain endocrine-disrupting chemicals has been shown to alter the development of the male rat mammary gland (Filgo et al., 2016; Kass et al., 2015; Mandrup et al., 2016). Some of them induce developmental changes only in the male mammary gland (You et al., 2002), whereas others induce alterations in both the male and female mammary gland, affecting males earlier than females (Mandrups et al., 2012). Based on these results, the mammary gland of male rats constitutes a useful model to study the effects of potential endocrine disruptors. Recently, we have shown that postnatal exposure to the endocrine disruptor endosulfan induces premalignant lesions in the mammary gland of post-pubertal male rats (Altamirano et al., 2017a), and that an early postnatal subcutaneous treatment with a GBH increases mammary gland development of pre-pubertal male rats and increases mast cell infiltration, proliferation index, and ESR1 expression in post-pubertal ones (Altamirano et al., 2018). Considering that both the susceptibility window and the administration route to which animals are exposed may have different impacts on the endocrine system (Bergman et al., 2012; Yang et al., 2015), in the present study, we decided to assess the effects of an oral exposure to a GBH during development (*in utero* + lactation). Gestational and neonatal stages are critical periods of mammary gland development, which could be altered by exposure to endocrine disruptors. Since the primary ducts are formed and begin their extension and branching during these stages,

any interference of these events could lead to altered mammary development, leaving lasting effects on the gland (Fenton, 2006). Given that, as mentioned above, glyphosate residues have been detected in water courses (Ronco et al., 2016), crops at harvest and food (Myers et al., 2016), as well as maternal and umbilical cord serum (Kongtip et al., 2017), the risk of oral and *in utero* exposure exists. Thus, the aims of the present study were to determine whether oral developmental exposure to a GBH affects mammary gland growth and development in pre- and post-pubertal male rats, and to evaluate the possible mechanisms involved.

2. Materials and methods

2.1. Animals

All the experimental protocols were approved by the Ethical Committee of the Facultad de Bioquímica y Ciencias Biológicas of the Universidad Nacional del Litoral (FBCB-UNL), Santa Fe, Argentina. Animals were treated humanely and with regard for alleviation of suffering. Sexually mature female and male rats (90 days old) of a Wistar-derived strain bred at the Department of Human Physiology (FBCB-UNL) were used. The animals were maintained in a controlled environment ($22 \pm 2^\circ\text{C}$; 14 h of light) and tap water was supplied *ad libitum* in glass bottles with rubber stoppers surrounded by a steel ring.

2.2. Experimental procedures

Pregnant rats (F0) on gestational day 9 (GD9) were weighed and randomly divided into three groups (8–10 dams/group): a) Control, fed with a standard diet (laboratory pellet chow, 16–014007 Rat-Mouse diet, Nutrición Animal, Santa Fe, Argentina); for more information regarding food composition see Andreoli et al. (2015); b) GBH3.5, fed with a standard diet supplemented with a GBH (Magnum Super II, Grupo Agros SA; Glyphosate 66.2% - Potassium Salt, acid equivalent 54%) in a dose of 3.7 mg of glyphosate/kg bw/day; and c) GBH350, fed with a standard diet supplemented with a GBH (Magnum Super II) in a dose of 352 mg of glyphosate/kg bw/day. The doses of glyphosate were selected based on the no-observed-adverse-effect level (NOAEL) of 1000 mg/kg bw/day for maternal and developmental toxicity established for this herbicide by the US EPA (Williams et al., 2000) and were in the order of magnitude and a 100-times lower than the NOAEL of 300 mg/kg bw/day specified by the EFSA (EFSA, 2015). As previously described by Milesi et al. (2018), the laboratory pellet chow was supplemented with GBH, considering the amount of active ingredient present in the formulation (54 g of glyphosate per 100 ml of GBH) and the average food consumption of F0 dams, and the oral treatments to F0 dams were administered from GD9 until weaning (lactation day 21 (LD21)) (Fig. 1). In all groups, the food was replaced three times per week and the pellet-based paste was prepared freshly the same day on which the food was changed. As reported by Milesi et al. (2018), glyphosate integrity was preserved during three consecutive days, which corresponds to the period of food replacement. Individual body weight and food intake of F0 dams were recorded three times per week throughout the treatment and the doses were calculated on the basis of their average body weight and food consumption during pregnancy and lactation. In addition, trunk blood samples of F0 mothers were collected at the end of the lactation period (LD21) for determination of serum levels of glyphosate and its metabolite AMPA. After delivery (postnatal day 0 - PND0), the pups were weighed and sexed, and litters of eight pups (four males and four females) were left with lactating mothers until weaning on PND21. Given that the GBH exposure of the dams was conducted in association with another research group of our Institute, which studies the effects of GBH on F1 and F2 female offspring, the *in vivo* parameters related to the experimental model (i.e. food intake) have already been published by them (see Milesi et al. (2018) for details).

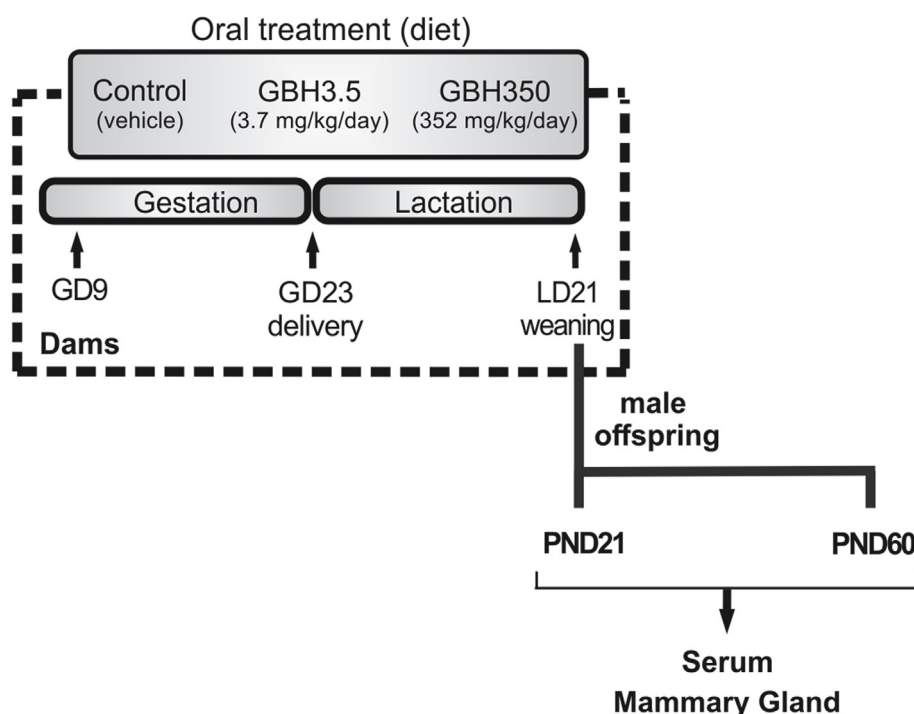


Fig. 1. Schematic representation of the experimental protocol used to study the effects of perinatal exposure to GBH on the development of the mammary gland of pre- and post-pubertal male rats. GD: gestational day, LD: lactation day, PND: postnatal day.

One male offspring per litter from each treatment group was sacrificed at either pre-puberty (PND21) or post-puberty (PND60), and both trunk blood and mammary gland samples were obtained. The male pups were weighed on PND21 and PND60, and the body weight gain was calculated as the difference between weight on each day (g) (PND21 or PND60) and the weight at delivery (g), divided by the weight at delivery (g). In addition, the onset of puberty was assessed by determining the day of preputial separation. Blood serum samples were stored at -80°C until E2, T and prolactin (PRL) levels were quantified. One abdominal-inguinal mammary gland chain was randomly chosen to be processed for whole mount analysis on both PNDs studied. On PND60, mammary gland N° 4 of the contralateral chain was fixed in 10% (v/v) buffered formalin for 6 h at room temperature and embedded in paraffin for histological examination and immunohistochemical assays (Kass et al., 2015), whereas mammary glands N° 5 and 6 were snap-frozen in liquid nitrogen and kept at -80°C for real-time RT-PCR (qRT-PCR) and DNA methylation analysis. The remaining males and all females were assigned to other experiments.

2.3. Determination of glyphosate and AMPA concentrations

Glyphosate and AMPA serum concentrations in F0 dams were measured on LD21 (at the end of the lactation period) by Ultra performance liquid chromatography - tandem mass spectrometry (UHPLC-MS/MS). The assays were performed at the PRINARC (Programa de Investigación y Análisis de Residuos y Contaminantes Químicos, Facultad de Ingeniería Química, UNL, Santa Fe, Argentina) and described by Milesi et al. (2018). The limit of detection (LOD) and quantification (LOQ) for glyphosate and AMPA were 1 and $2\mu\text{g L}^{-1}$, respectively.

2.4. Determination of hormones

Serum E2 and T levels in male offspring were determined on PND21 and PND60 by chemiluminescence assays by using an Immulite® 2000 system (Siemens Healthcare SA, Argentina), as described in Altamirano et al. (2017a). On PND60, PRL serum concentrations were assessed by

radioimmunoassay (Lacau-Mengido et al., 1996), at the Instituto de Biología y Medicina Experimental (IBYME, Buenos Aires, Argentina). The LODs of the assays were 15 pg/ml for E2, 0.15 ng/ml for T and 0.3 ng/ml for PRL, and the intra-assay coefficients of variation were 6.7% for E2, 10.1% for T and 7.8% for PRL.

2.5. Whole mount evaluation

Images of mammary gland whole mounts (WMs) were collected on PND21 and PND60, as previously described by Altamirano et al. (2017a). All images were analyzed with ImageJ software (NIH, U.S.A.; <https://imagej.nih.gov/ij>). The unequal illumination (shading correction) was corrected and the measurement system was calibrated with a reference slide. All the evaluations were carried out blinded to the treatment group on mammary gland N° 4. Different mammary gland parameters, including total area, perimeter, distance between mammary gland N° 4 and mammary gland N° 5, longitudinal growth, and number of terminal structures, were analyzed on mammary gland WM images, as previously described in Kass et al. (2015) and Altamirano et al. (2017a). Besides, to evaluate mammary gland development, the relative epithelial area was determined on PND21 and a triple blinded score was determined on PND60. The relative epithelial area was defined as the area occupied by the epithelial tree divided by the total area. Briefly, the area around the glandular epithelium was cleared from the image, the color channels were separated, the noise was removed, and finally, the epithelial tree area was assessed (Stanko et al., 2015). On PND60, mammary gland WMs were assigned a qualitative score on a 1–3 subjective scale, where a value of 1 represented poor mammary gland tree development and 3, the highest level of development. The qualitative scoring criterion was based on branching density, appearance of budding from ductal tree and lobuloalveolar development. Each gland was scored by three individuals blinded to treatment and averaged to obtain a final score.

2.6. Histological analysis

For histological examination, paraffin sections ($5\mu\text{m}$) of mammary

gland samples collected on PND60 were cut and stained with Sirius Red (Direct Red 80, Sigma-Aldrich, Argentina) in picric acid solution (picric acid) counterstained with Harris hematoxylin (Kass et al., 2001). The development of ducts and lobuloalveolar structures as well as the presence of mammary gland lesions (Durando et al., 2007; Mandrup et al., 2015) were evaluated using an Olympus BH2 light microscope (Olympus Optical Co., Ltd., Japan). In addition, the relative epithelial area (area occupied by epithelial structures/total mammary gland area *100) and the adipocyte's individual area was measured in 200× magnification images; at least 15 images per section and two sections (30 µm apart from each other) per mammary gland were evaluated. The average area of individual adipocytes was determined as described by Parlee et al. (2014). For both histological analyses, images were recorded using a digital camera (C-5000Zoom, Olympus), and measurements were performed using the ImageJ software (NIH, USA; <https://imagej.nih.gov/ij/>).

2.7. Proliferation and expression of steroid hormone receptors

Consecutive 5-µm sections were immunostained to evaluate the proliferation index (Ki67) and the expression of ESR1 and androgen (AR) on PND60. Immunoperoxidase staining was performed as previously described (Kass et al., 2015). Sections were incubated overnight at 4 °C with primary antibodies against Ki67 (Dako, Glostrup, Denmark), ESR1 (clone 6-F11, Novocastra Laboratories Ltd., Newcastle Upon Tyne, UK) and AR (N-20, Santa Cruz Biotechnology Inc., Santa Cruz, USA). Anti-rabbit (Zymed Laboratories Inc., CA, USA) or anti-mouse (Instituto de Salud y Ambiente del Litoral, Santa Fe, Argentina) secondary antibodies (biotin-conjugated) were used, and reactions were developed using an avidin-biotin peroxidase method with diaminobenzidine (Sigma-Aldrich) as a chromogen substrate. Each run included negative controls in which the primary antibody was replaced with non-immune mouse or rabbit serum (Sigma-Aldrich). Two mammary tissue sections per animal and at least 2000 cells per tissue section were analyzed. The percentages of Ki67, ESR1 and AR expression in the epithelial cells were quantified in each tissue section.

2.8. mRNA levels of ESR1, AR and ESR1 alternative transcripts

The mammary glands from 6 to 8 animals from each experimental group (Control, GBH3.5 and GBH350) obtained on PND60 were individually homogenized in TRIzol (Invitrogen, Buenos Aires, Argentina), and RNA was prepared according to the manufacturer's protocol. The concentration of total RNA was assessed by measuring the A260 value, and RNA was stored at –80 °C until needed. Then, 1 µg of total RNA was reverse-transcribed into cDNA following the method described in Altamirano et al. (2017b). Samples were analyzed in triplicate, and a sample without reverse transcriptase was included to detect contamination by genomic DNA. An optimized real-time PCR protocol was used to analyze the mRNA expression levels of ESR1, AR and ESR1 transcripts containing the 5' untranslated exons OS, ON, O, OT, and E1. The primer sequences designed for cDNA amplification are shown in Table S1 (Supplementary data). cDNA levels were determined by using a real-time PCR system StepOne Cyclor (Applied Biosystems Inc., Life Technologies, Carlsbad, USA). A total of 5 µl of cDNA was combined with HOT FIRE Pol EvaGreen qPCR Mix Plus (Solis BioDyne; Biocientifica, Rosario, Argentina) and 10 pmol of each primer (Invitrogen) at a final volume of 20 µl. After initial denaturation at 95 °C for 15 min, the reaction mixture was subjected to successive cycles of denaturation at 95 °C for 15 s, annealing at 57 °C (AR and ESR1-ON and E1), 58 °C (ESR1-OS and O), 59 °C (ESR1-OT) and 60 °C (ESR1 and L19) for 15 s, and extension at 72 °C for 15 s. Product purity was confirmed by dissociation curves, and random samples were subjected to agarose gel electrophoresis. Negative DNA template controls were included in all the assays and yielded no consistent amplification. The threshold cycles (Ct) and PCR efficiency were calculated using the Step One

software (Applied Biosystems Inc.). No significant differences in the Ct values of L19 were observed between the different experimental groups. The relative expression level of each target was calculated using the standard curve method (Cikos et al., 2007) and L19 was used to normalize the Ct values.

2.9. Bioinformatics

The ESR1 promoter regions were analyzed for CpG islands by using the Sequence Manipulation Suite program (<http://www.bioinformatics.org/sms2>). A CpG island was defined as a 200-bp DNA sequence with a calculated percentage of CpGs of more than 50% and a calculated-versus-expected CpG distribution higher than 0.60. These regions were also checked for restriction sites for *Bst*UI and *Mae*II to evaluate the number of methylation-sensitive sites. PCR primers were designed with the software Vector NTI Suite Version 6.0 (Infomax Inc., North Bethesda, MD, USA).

2.10. DNA methylation-sensitive analysis

The DNA methylation status of the ESR1 promoters was analyzed using a combination of single digestions with methylation-sensitive restriction enzymes and subsequently performing real-time PCR analysis (Altamirano et al., 2017b; Milesi et al., 2017). Genomic DNA was isolated from mammary gland samples by using phenol/chloroform/isoamyl alcohol extraction. The total concentration of DNA was quantified by A280 and stored at 2–8 °C until needed. Equal quantities (1 µg) of total DNA were digested with five units of *Hind*III (Promega, Madison, USA) to reduce the size of the DNA fragments and then purified with phenol/chloroform extraction and ethanol precipitation. Then, 1 µg of *Hind*III-cleaved DNA was incubated for 1 h with one unit of *Mae*II (Roche Applied Science, IN, USA) or 10 units of *Bst*UI (New England BioLab, Beverly, MA, USA) and 1X enzyme buffer at 50 °C or 60 °C respectively, following the manufacturer's instructions. The digestion products were purified with the phenol/chloroform method. The relative expression level of the different DNA regions was analyzed by real-time PCR (as mentioned above). The primer sequences are shown in Table S2 (Supplementary data). Four pooled samples from each group were quantified in triplicate. The methylation restriction enzymes *Mae*II or *Bst*UI are unable to cut at methylated sites, allowing amplification of the fragment. A region without *Mae*II or *Bst*UI restriction sites was used as an internal control. The relative degree of methylation was determined by plotting Ct values against the log input (internal control), which yielded standard curves for the quantification of unknown samples (Cikos et al., 2007).

2.11. Statistical analysis

All data are expressed as the mean ± SEM. An exploratory analysis was first conducted to evaluate whether the data were normally distributed (Shapiro–Wilk test) and variances were homogeneous (Levene's test); then, the appropriate statistical analysis was selected. WM parameters (total area, perimeter, distance between mammary gland N° 4 and mammary gland N° 5 and longitudinal growth) were evaluated using a linear model with treatment and weight as independent variables (response variable ~ treatment + weight) to analyze whether the body weight had a significant effect on these response variables (Altamirano et al., 2018). After discarding the effect of body weight, WM determinations as well as food intake, body weight gain, histological analysis and protein expression were evaluated using ANOVA, to obtain the overall significance, and Dunnett's post-hoc test was used to compare each experimental group with the control group. The E2, T and PRL serum levels, the number of terminal end buds (TEBs), and the WM score in the control and both GBH-treated groups were analyzed using Kruskal–Wallis test followed by Dunn's method for multiple comparisons. For qRT-PCR and DNA methylation analysis, the

Mann-Whitney *U* test was used. Values with $p < 0.05$ were accepted as significant. Analyses were carried out using R software (The R Foundation for Statistical Computing, <http://www.r-project.org/>).

3. Results

As previously described by Milesi et al. (2018), there were no changes in the average body weight or food consumption of F0 dams, and glyphosate serum concentrations were 0.039 ± 0.006 mg/l in GBH3.5-exposed dams and 3.80 ± 1.20 mg/l in GBH350 ones, whereas glyphosate was not detectable in control F0 dams. In addition, no AMPA was detected in the F0 dams of any of the groups studied. Also, the treatment with GBH from GD9 until weaning, administered through the diet, produced no signs of embryo toxicity (the number of live-born pups per litter was similar among groups), and no abnormal maternal or nursing behaviors. The length of gestation was unaltered, and no gross malformations were observed in the F1 pups at delivery or weaning (PND21). The litter sex ratio showed no alterations and was within the normal range (50% females and 50% males).

The developmental exposure to GBH induced no changes in the body weight of male pups at delivery or in the body weight gain at weaning and PND60 (Table 1). The onset of puberty, assessed by the day of preputial separation (data not shown), was not affected by exposure to GBH; all male offspring reached puberty by PND35. E2 and T serum levels were similar between groups on both PNDs studied (Table S3, supplementary data), whereas PRL serum concentration was slightly increased in GBH350 animals compared with control rats on PND60 (Control: 3.3 ± 1.3 ng/ml vs GBH3.5: 3.8 ± 1.7 ng/ml vs GBH350: 8.0 ± 2.9 ng/ml).

3.1. Mammary gland histoarchitecture of pre- and post-pubertal animals

Mammary gland WMs from pre- and post-pubertal male animals were analyzed to assess mammary growth and development (Table 2). Representative images of WM samples are shown in Fig. 2A–F. On PND21, the mammary gland from all samples reached the lymph node in all experimental groups. Also, the presence of TEBs and alveolar buds was evident, and all the mammary gland parameters evaluated in the WMs were similar between GBH-treated and control groups. As previously reported (Altamirano et al., 2017a), on PND60, mammary gland structures grew past the lymph node, the 4th and 5th glands converged, the lateral growth was increased, reaching the edge of the fat pad, and male mature structures were developed. At this age, GBH3.5 treatment reduced the total area and the perimeter of the gland and induced the persistence of TEBs, all of which suggests a delay in mammary gland growth. On the other hand, animals exposed to GBH350 presented higher longitudinal growth and a less developed mammary gland tree than control males.

Taking into consideration that GBH treatment modified the mammary gland growth only on PND60, a histological analysis was

Table 1

Reproductive parameters in GBH-exposed dams and body weight of male offspring.

	Control	GBH3.5	GBH350
% successful pregnancies	100	100	83
No. of pups/litter	9.90 ± 0.90	10.80 ± 0.60	9.20 ± 0.50
F1 ♂ weight at delivery (g)	5.89 ± 0.09	5.59 ± 0.09	5.87 ± 0.13
F1 ♂ weight gain on PND21	4.89 ± 0.17	4.77 ± 0.17	4.93 ± 0.15
F1 ♂ weight gain on PND60	36.45 ± 0.79	34.62 ± 0.86	38.78 ± 0.92

Values are expressed as mean $\pm$ SEM of 8–10 either dams/group or male rats/group.

Weight gain was defined as the difference between male offspring weight (g) on PND21 or PND60 and the weight at delivery (g), divided by the weight at delivery (g).

Table 2

Whole mount analysis of GBH-exposed male rats.

	Control	GBH3.5	GBH350
PND21			
Total area (mm ²)	26.7 ± 1.7	25.4 ± 1.9	24.2 ± 2.1
Perimeter (mm)	27.2 ± 1.0	28.2 ± 1.7	27.1 ± 1.4
Distance between MG No. 4 and MG No. 5 (mm)	1.8 ± 0.3	1.3 ± 0.2	1.4 ± 0.4
TEBs (No.)	1.8 ± 0.7	0.6 ± 0.2	2.6 ± 0.9
Relative epithelial area (%)	29.9 ± 1.2	25.8 ± 1.9	30.0 ± 1.6
PND60			
Total area (mm ²)	493.2 ± 25.3	388.2 ± 13.9^a	524.9 ± 18.2
Perimeter (mm)	94.1 ± 3.1	84.2 ± 1.5^a	99.3 ± 2.3
Longitudinal growth (mm)	7.1 ± 0.4	6.4 ± 0.6	9.4 ± 0.6^a
TEBs (No.)	0.5 ± 0.2	2.3 ± 0.7^b	0.3 ± 0.2
Mammary gland tree development (score)	2.6 ± 0.1	2.7 ± 0.1	1.4 ± 0.1^a

Values are expressed as mean $\pm$ SEM of 8–10 male rats/group/day.

MG: mammary gland; TEBs: terminal end buds.

^a $p < 0.05$ (One-way ANOVA, followed by Dunnett's post-hoc test).

^b $p < 0.05$ (Kruskal Wallis, followed by Dunn's post-hoc test).

performed in post-pubertal males. Representative images are shown in Fig. 2G–L. On PND60, the mammary gland of control animals presented the lobuloalveolar pattern characteristic of adult male rats (Filgo et al., 2016; Lucas et al., 2007). Also, the lobular structures exhibited foamy and vacuolated epithelial cells arranged around indistinct lumen (Fig. 2G, J). GBH3.5-exposed animals showed epithelial development similar to that of control animals (Fig. 2H, K), whereas GBH350-exposed males showed less epithelial development than the control group, mostly of alveolar structures (Fig. 2I, L). In accordance with these features, the relative epithelial area (%) decreased in the GBH350 group compared to the control and no differences were observed in animals treated with the lower dose (Control: 25.83 ± 1.59 , GBH3.5: 28.79 ± 2.01 , GBH350: $19.81 \pm 1.90^*$; $*p < 0.05$). In addition, an increase in adipose tissue development, which was accompanied by an increased adipocyte area (Control: $500.3 \pm 48.3 \mu\text{m}^2$, GBH3.5: $477.3 \pm 29.6 \mu\text{m}^2$, GBH350: $1053 \pm 140.9^* \mu\text{m}^2$; $*p < 0.05$), was observed in GBH350-exposed males.

3.2. Proliferation and expression of steroid hormone receptors in the mammary gland of post-pubertal male rats

To evaluate whether the morphological alterations observed in the male mammary gland after developmental exposure to GBH were related to changes in the proliferation and mammary gland steroid hormone receptor expression, the proliferation index (percentage of Ki67-positive cells) and ESR1 and AR protein and mRNA expression were assessed on PND60. Representative images of mammary epithelial cells expressing Ki67, ESR1 and AR are shown in Fig. 3A. In GBH3.5-exposed animals, the proliferation index was similar to that in the control ones, but ESR1 protein and mRNA expression levels were lower. In addition, these changes were accompanied by an increase in AR protein expression and a decrease in AR mRNA levels. Contrary to that observed in GBH3.5-exposed rats, in GBH350-exposed rats the proliferation index was lower than in control rats. Also, ESR1 protein and mRNA expression levels were reduced, whereas AR expression was not modified by the exposure to GBH350 (Fig. 3B and C).

3.3. Relative abundance of ESR1 alternative transcripts in post-pubertal animals

To determine whether the decrease in total ESR1 mRNA expression was associated with changes in transcriptional promoter usage, the relative expression levels of all exons encoding 5'UTR of the rat ESR1 gene were studied using qRT-PCR. A schematic representation of the

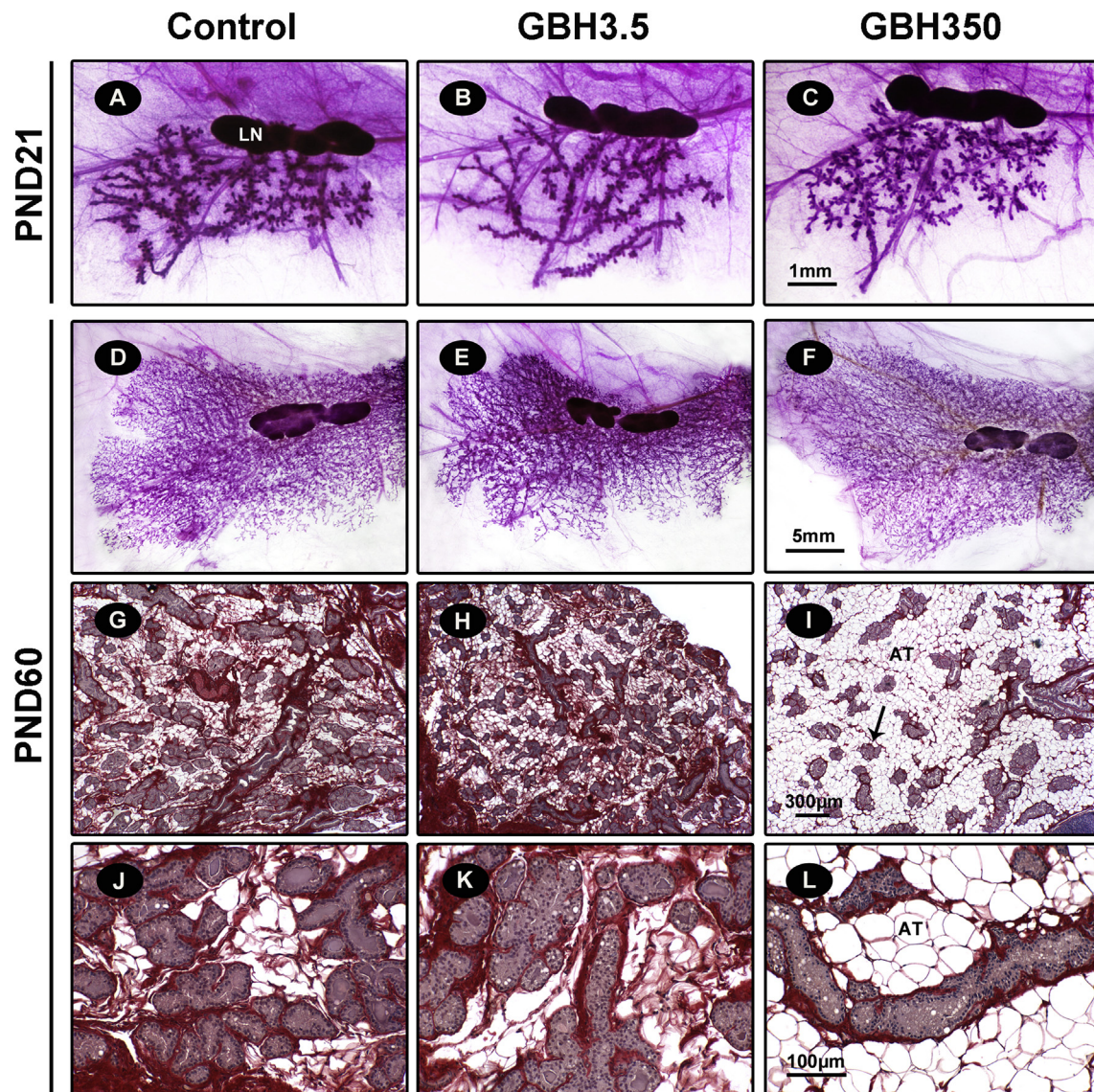


Fig. 2. Representative images of whole mounts (WMs) (A–F) on PND21 and PND60 and picosirius-hematoxylin-stained sections (G–L) of mammary glands from post-pubertal males. On PND21, no differences were observed between GBH-treated rats and control ones, whereas on PND60, males exposed to GBH3.5 showed a reduced area but mammary gland histology (E, H and K) similar to that of control males. GBH350-exposed animals presented a higher longitudinal growth and decreased mammary tree development than control rats (F). The histological sections of GBH350-treated rats (I, L) show a reduced relative epithelial area, accompanied by a lower development of lobuloalveolar structures (I, arrow) and a prominent adipose tissue (I and L). LN: lymph nodes; AT: adipose tissue. Images in each line present the same magnification.

genomic organization of the promoter region of the rat *ESR1* gene is presented in Fig. 4A. Regardless of the experimental group, *ESR1* gene transcription in the male rat mammary gland was regulated by the promoters associated with the exons OS, O, OT and E1, whereas no expression of transcripts containing the ON exon was detected under the experimental conditions of the study. Nevertheless, both doses of GBH altered the abundance of transcript variants. GBH3.5-exposed animals presented a lower expression of the *ESR1*-OS variant, whereas GBH350-exposed ones presented a lower expression of the other three transcript variants (*ESR1*-O, OT and E1) than control rats (Fig. 4B).

3.4. Methylation status of *ESR1* transcriptional promoters

Considering the changes induced by the GBH treatment in the expression of *ESR1* transcript variants, the methylation status of the transcriptionally active promoters and surrounding DNA areas was determined to assess whether these modifications were related to

differential DNA methylation. To search for potential sites for DNA methylation, *ESR1*-OS, O, OT and E1 promoter regions were analyzed for CpG islands and restriction sites for *Bst*UI or *Mae*II (Milesi et al., 2017). CpG islands and restriction sites for *Bst*UI or *Mae*II were identified in all the promoters studied. Fig. 5 shows the scheme of OS, O, OT and E1 associated promoters, CpG islands and sites studied by qRT-PCR.

The methylation status was analyzed in all the promoters of the transcript variants. In GBH3.5-exposed rats, only the *Bst*UI site in the OS promoter presented an increased methylation status, whereas, in GBH350-exposed rats, the *Mae*II and the two *Bst*UI sites in the O promoter, the *Bst*UI site in the OT promoter, and the *Bst*UI site in the E1 promoter showed higher methylation status than the control group (Fig. 5). In all cases, the reduced mRNA expression of the transcript variants was associated with the hypermethylation of their promoters.

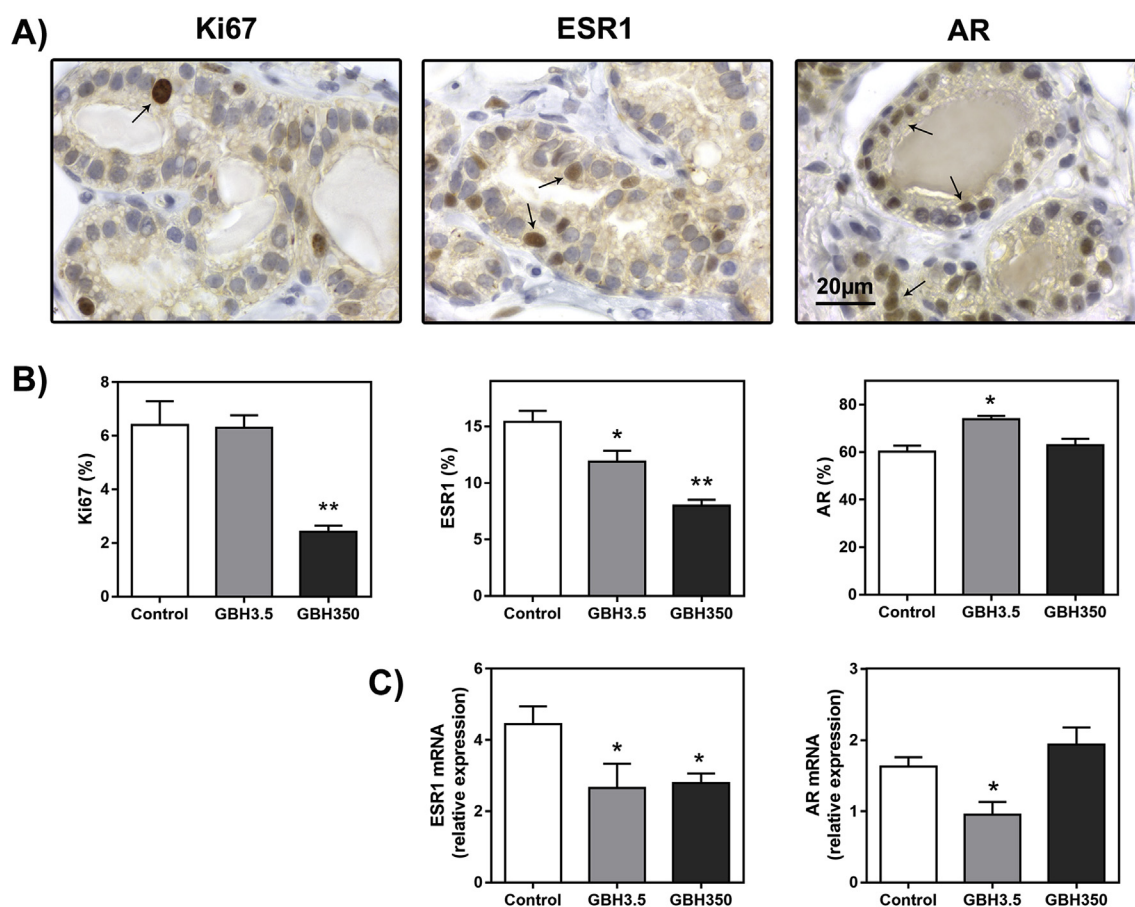


Fig. 3. Proliferation index and steroid hormone receptor expression in post-pubertal male rats. **(A)** Representative images of Ki67 (proliferation) and ESR1 and AR protein expression in control animals on PND60. Arrows indicate positive cells; all images have the same magnification. **(B)** Percentage of positive cells expressing Ki67, ESR1 and AR protein. Bars represent the mean values ± SEM of 6–8 male rats/group (*p < 0.05; ANOVA followed by Dunnett's post-test). **(C)** ESR1 and AR mRNA relative expression. Bars represent the mean values ± SEM of 6–8 males/group and the samples were normalized to L19 expression (*p < 0.05, Mann Whitney U test).

4. Discussion

Currently, due to the increasing and widespread use of glyphosate formulations (Mesnage et al., 2015; Vandenberg et al., 2017) and their potential endocrine-disrupting properties (Cai et al., 2017; Guerrero Schimpf et al., 2017; Ingaramo et al., 2017; Varayoud et al., 2017), their possible consequences on human health are an issue of concern. In the present study, the effect of developmental exposure to a GBH was evaluated on the male rat mammary gland, given the high sensitivity of this organ to endocrine-disrupting chemicals (Altamirano et al., 2017a, 2018; Andrews et al., 2002; Kass et al., 2015; Miousse et al., 2013). Developmental exposure to GBH resulted in altered development of the mammary gland of post-pubertal males but in no changes in the mammary gland of pre-pubertal ones. On PND60, the mammary gland of males treated with GBH3.5 showed a decrease in its growth and an increased number of TEBs, but its morphology was similar to that of the control group. On the other hand, the mammary gland of GBH350-exposed animals exhibited greater longitudinal growth but less mammary tree development, accompanied by lower epithelial area and higher adipocyte tissue development.

The glyphosate serum concentrations obtained in F0 dams were similar to those reported in toxicokinetic studies for a single oral dose of glyphosate (Anadon et al., 2009). In addition, the serum values of glyphosate in dams exposed to GBH3.5 (0.039 ± 0.006 mg/l) were within the range of those found in pregnant women at childbirth in Thailand (Kongtip et al., 2017) (0.0002 – 0.1891 mg/l), whereas those in GBH350-exposed dams (3.80 ± 1.20 mg/l) were 20 times higher than

the maximum value obtained by Kongtip et al. (2017). As previously reported by Anadon et al. (2009), in rats, glyphosate is poorly metabolized and AMPA is the main metabolite; however, in the present study, AMPA was not detected in the serum of F0 dams. It is possible that glyphosate metabolism in lactating dams differs from that in adult male rats, as those used by Anadon et al. (2009), as it is well known that drug metabolism shows sex-based differences (Czerniak, 2001).

In a previous study, we showed that direct subcutaneous exposure to GBH in the first week of life enhances mammary growth and increases the number of TEBs in pre-pubertal males (Altamirano et al., 2018). In contrast, in the present study, perinatal oral exposure of pregnant dams to GBH and, therefore, indirect treatment of male offspring through maternal exposure, caused no alterations in any of the parameters of mammary gland development evaluated on PND21, at either dose studied. These different responses could be due to alternative routes and/or windows of exposure (Yang et al., 2015). Although treatment with certain endocrine disruptors has been shown to cause abnormal mammary development in pre-pubertal male rats (Altamirano et al., 2017a; Filgo et al., 2016; Kass et al., 2015), it has been reported that early exposure to this kind of compounds could cause little observable effects that may not arise until the onset of puberty or adulthood (Fenton, 2006; Schug et al., 2011). On PND60, GBH3.5-treated animals presented a smaller area and perimeter of the mammary gland and a higher number of TEBs than control ones. Considering that TEBs are undifferentiated proliferative structures that are highly susceptible to an oncogenic insult, an increase in the number of TEBs or the persistence of these immature structures in exposed

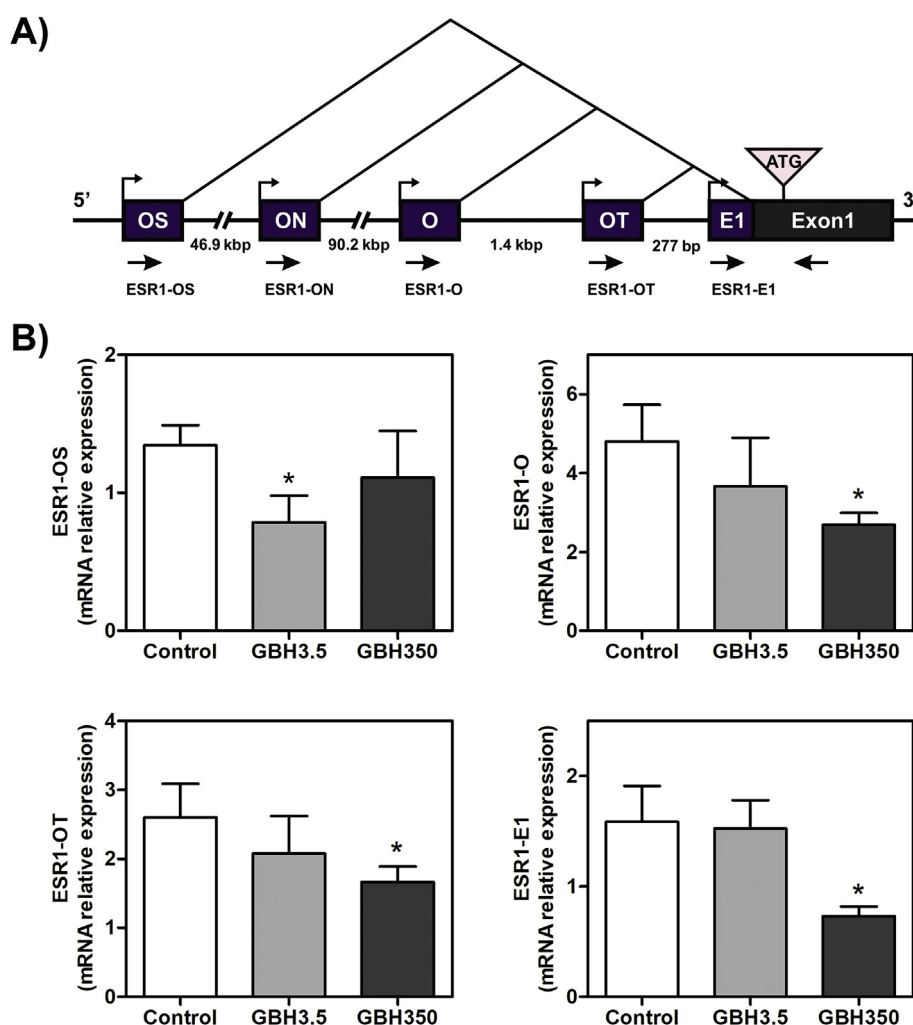


Fig. 4. ESR1 transcriptional variants. **(A)** Genomic organization of the promoter region of the rat ESR1 gene. Relative positions and orientations of promoter-specific PCR primers are indicated by black arrows and the gray box shows the region common to all the alternative 5'UTRs of ESR1. ATG: translation initiation site. **(B)** mRNA relative expression of ESR1 transcripts with alternative 5'UTR regions. Bars represent the mean values $\pm$ SEM of 6–8 males/group and the samples were normalized to L19 expression (* p < 0.05, Mann-Whitney U test).

males could increase the sensitivity of the gland to endogenous or exogenous perturbations (Fenton, 2006; Paine and Lewis, 2017). In addition, the changes observed in mammary gland WMs suggest a delay in the mammary growth of GBH3.5-exposed males, where mammary ductal elongation has not finished yet. Further analysis at previous and later time points would be necessary to confirm this hypothesis. Regarding GBH350-exposed animals, they presented higher longitudinal growth and decreased mammary tree development. It is known that TEBs are primarily responsible for epithelial extension and that, in males, they begin to differentiate into terminal ends before reaching the end of the fat pad (Filgo et al., 2016). Although the number of TEBs in control and GBH350-treated animals at the time points analyzed was similar, TEBs could have differentiated later than in control animals and could have caused more elongated ducts.

In the present study, the histological features and proliferation index of the mammary gland of GBH3.5-exposed males were similar to those of the mammary gland of control rats. In contrast, in GBH350-exposed males, the mammary gland relative epithelial area was smaller (in accordance with the reduced development found in the WMs), the development of lobuloalveolar structures characteristic of males was decreased, and the adipose tissue became more prominent. These features partly resemble a female mammary gland (Lucas et al., 2007). Taking into consideration that hyperprolactinemia can induce mammary gland feminization in male rats (Cardy, 1991; Lucas et al., 2007),

the trend to increased PRL serum levels observed in GBH350-exposed animals could explain these histological changes.

The morphological changes observed in GBH350-treated animals were accompanied by a decreased epithelial proliferation and down-regulation of ESR1 protein expression, without alteration of AR protein levels. It has been described that the post-pubertal male mammary gland expresses ESR1 (Altamirano et al., 2017a; Filgo et al., 2016), but, in contrast to that occurring in females, the regulation of ESR1 expression as well as the effects of estrogens on the male gland have not been completely elucidated. Glyphosate has been found to induce ESR1 expression in human breast cancer cells *in vitro* (Thongprakashsang et al., 2013) as well as in the male rat mammary gland *in vivo* (Altamirano et al., 2018). In contrast, in the present study, ESR1 expression was decreased. In this sense, it needs to be considered that, here, ESR1 expression was evaluated long after the exposure had ended and that the administration route and timing of exposure were different from those used in Altamirano et al. (2018). Since there were no differences in E2 or T serum levels between groups, the reduced expression of ESR1 may alter the normal mammary response to steroid hormones at puberty and affect the lobuloalveolar development characteristic of post-pubertal males. In this regard, lobuloalveolar atrophy has been observed in the mammary gland of male rats treated with tamoxifen (Kennel et al., 2003), a classical estrogen antagonist in the mammary gland.

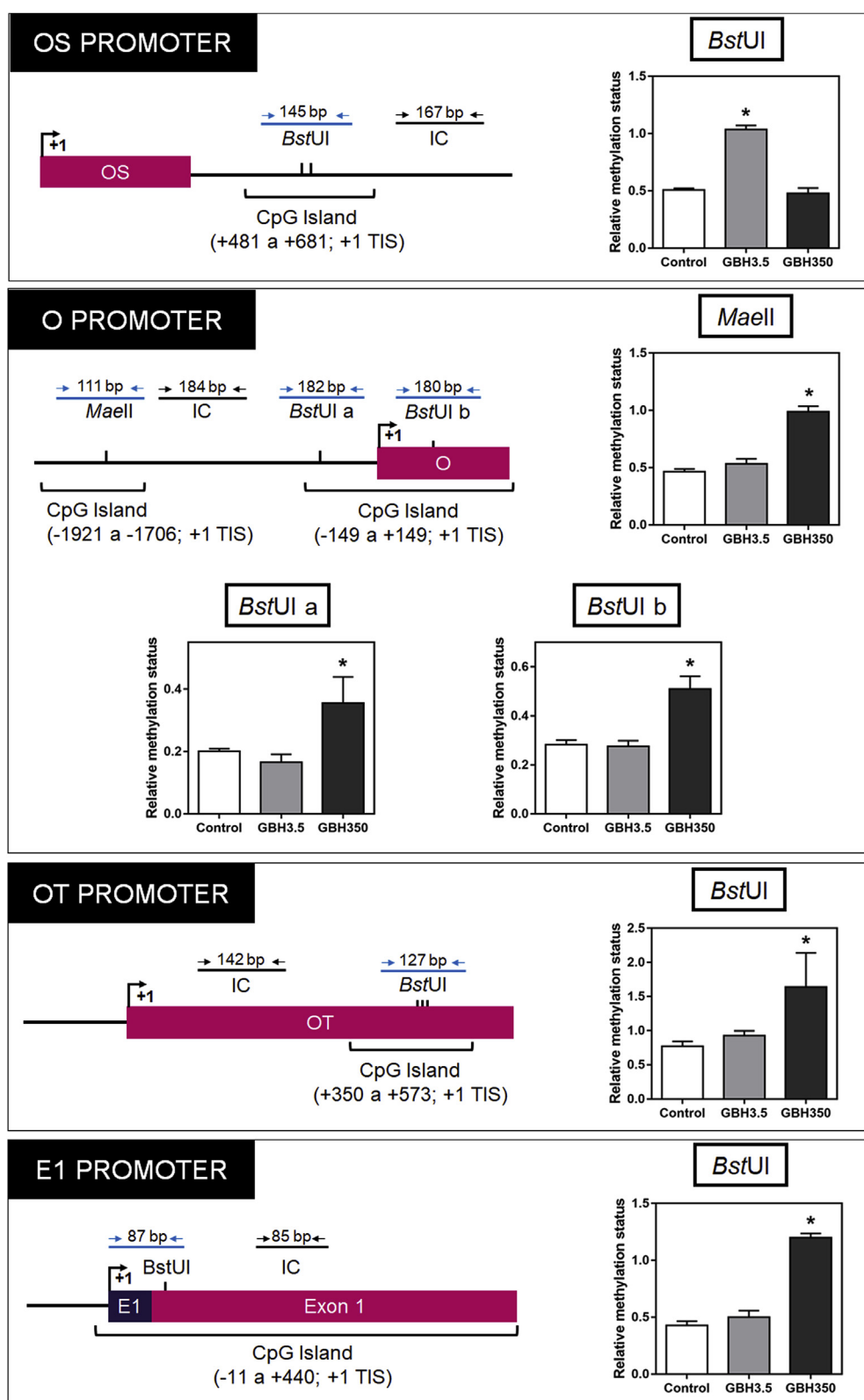


Fig. 5. Methylation status of the ESR1 promoter region. Maps of ESR1-OS, -O, -OT and -E1 promoters and their CpG islands are shown, and target sites for digestion by *Bst*UI and *Mae*II methylation-sensitive restriction enzymes are indicated. Positions and orientations of PCR primers are indicated by arrows. +1: transcription initiation site; IC: internal control region. Bars represent the relative methylation status of the ESR1 promoters and values are expressed as the mean $\pm$ SEM of four pooled samples quantified in triplicate (* $p < 0.05$, Mann Whitney U test).

The GBH-induced effects on ESR1 protein expression could be mediated by a reduction of ESR1 transcription, as ESR1 mRNA levels were decreased by the developmental exposure to GBH. Furthermore,

the ESR1 transcription induced by three different promoters (ESR1-O, OT and E1) was affected in GBH350-exposed animals. It has been described that ESR1 expression occurs through different promoters

depending on the tissue and physiological or developmental stages (Kato et al., 1998; Parodi et al., 2015) and that, in agreement with our findings, exposure to different chemical compounds could alter the relative abundance of the alternative ESR1 transcripts and therefore modulate ESR1 expression (Milesi et al., 2017; Monje et al., 2007; Parodi et al., 2015). DNA methylation, one of the best-known epigenetic modifications, plays a key role in mammalian development (Smith and Meissner, 2013), and a perturbation of its pattern is frequently observed in disease (Jones, 2012; Schubeler, 2015). Both the loss of expression of ESR1, which is frequently observed in breast cancer, and the presence of triple negative tumors are often associated with hypermethylation of ESR1 regulatory regions (Hervouet et al., 2013). In our research, analysis of the methylation status of ESR1 showed that, in GBH350-exposed males, the reduced mRNA expression of ESR1-O, -OT and -E1 was correlated with the hypermethylation of the CpG islands of their promoters. Related to this, it has been published that exposure to endocrine disruptors during sensitive periods of development can perturb epigenetic programming (Jacobs et al., 2017). Bisphenol A (BPA) exposure alters DNA methylation in regulatory regions of the β -Casein gene in the lactating mammary gland (Altamirano et al., 2017b) and generates hypermethylation of the promoter region of ESR1 in the testis (Doshi et al., 2011). Furthermore, the uterus of female rats treated with endosulfan presents decreased DNA methylation levels in ESR1 regulatory regions that are associated with the up-regulation of its transcription (Milesi et al., 2017).

As mentioned before, the mammary gland morphology and proliferation index of GBH3.5-exposed males were similar to those of control rats on PND60. This was accompanied by a reduction of ESR1 mRNA and protein expression. Nevertheless, unlike that observed with the higher dose, only the transcriptional variant ESR1-OS was modified and presented hypermethylation of its regulatory region, compared to the control. Interestingly, in animals treated with GBH3.5, AR protein expression was increased while mRNA levels were decreased. These results could reflect differences in the translation efficiency or protein stability; in this sense, it has been reported that diverse post-transcriptional mechanisms modulate AR protein levels (Burnstein, 2005; Lee and Chang, 2003). Although some previous studies in male rats exposed to GBH described alterations in T serum levels, testicular morphology and function (Romano et al., 2010, 2012) and mammary gland development, others found no differences in AR expression (Altamirano et al., 2018; Cassault-Meyer et al., 2014; Dai et al., 2016). Considering these results, further experiments would be necessary to clarify the effects of GBH on AR expression and the possible mechanisms involved. In the present study, although ESR1 expression was decreased in both GBH3.5 and GBH350-treated animals, the mammary gland morphology was not affected in GBH3.5-treated animals. Regarding this, the decrease in ESR1 expression was less marked in males exposed to the lower dose, which may explain the absence of changes in the morphology of the mammary gland of GBH3.5-exposed males.

Taken together, our findings show that GBH exposure during the developmental period alters mammary gland development and steroid hormone receptor expression and that these changes involve epigenetic mechanisms. However, it is necessary to note that the described effects of GBH exposure could be due to the active principle (glyphosate), the co-formulants or both acting together. Several studies have shown that GBH-commercial formulations are more toxic than glyphosate alone (Mesnage and Antoniou, 2018; Mesnage et al., 2015; Myers et al., 2016) and further research is needed to clarify whether glyphosate, the adjuvants or its combination are responsible for the observed effects on male mammary gland development. Finally, the results of the present study suggest an endocrine disruption effect of GBH *in vivo* and highlight the importance of evaluating different administration routes, timings of exposure, and time points, since the effects on the mammary gland and the moment at which they arise can be different.

Conflicts of interest

The authors declare that there are no conflicts of interest that could be perceived as prejudicing the impartiality of the research reported.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mce.2018.11.005>.

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